Image Classification with Pretrained CNNs

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1 Introduction

Convolutional Neural Networks (CNNs) have achieved remarkable performance on large-scale image classification tasks such as ImageNet. In this task, you will work on a **10-class classification problem** using subsets of ImageNet. You are required to fine-tune pretrained models on this dataset, modify the classification head, and evaluate the performance of your models. The goal is to gain hands-on experience with transfer learning, model fine-tuning, and evaluation of deep learning models.

2 Dataset Selection and Preparation

The datasets for this task will be provided in a shared drive.

Dataset folder contains three subfolders:

- `train/` – Training images organized into class-wise folders.
- `test/` – Test images organized into class-wise folders.

You are required to write a **PyTorch DataLoader** to load the dataset. Make sure to:

1. Apply appropriate preprocessing and normalization (e.g., resize to 224×224, normalize with ImageNet mean and standard deviation).
2. (Optional) Use data augmentation (e.g., random cropping, flipping) for the training set.
3. (Optional) Use center cropping or resizing for validation and test sets (no augmentation).
4. Ensure that the DataLoader returns images and labels correctly for all three splits.

Your code should clearly demonstrate how the dataset is loaded and prepared for training and evaluation.

3 Experiments

You are expected to carry out the following experiments:

1. Use the following pretrained models from PyTorch:
    - ResNet-18
    - ResNet-34

- VGG-19

2. Replace the classification head of each model to adapt it to the 10-class dataset.
3. Train each model on the provided training set and validate using the validation split.
4. Optionally you can try different hyperparameter tuning strategies (e.g., learning rate scheduling, optimizers, regularization).
5. Evaluate the models on the test set and report the test accuracy as the main performance metric.